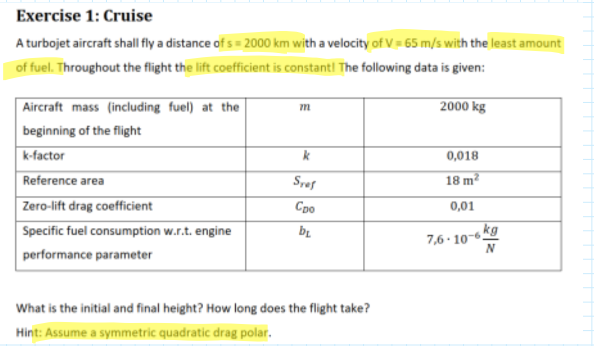
Exercise 7

Montag, 24. August 2026 09:30



Exercise 1: Cruise

A turbojet aircraft shall fly a distance of s = 2000 km with a velocity of V = 65 m/s with the least amount of fuel. Throughout the flight the lift coefficient is constant! The following data is given:

Aircraft mass (including fuel) at the beginning of the flight	m	2000 kg
k-factor	k	0,018
Reference area	S_{ref}	18 m^2
Zero-lift drag coefficient	C_{D0}	0,01
Specific fuel consumption w.r.t. engine performance parameter	b_L	$7{,}6\cdot10^{-6}\,\frac{kg}{N}$

What is the initial and final height? How long does the flight take?

Hint: Assume a symmetric quadratic drag polar.

	V	C_L	ρ
Type I	const	const	/
Type II	const	/	const.

negative

→ Select the distance equation: $S = -\int \frac{V^{n_V+1}}{g\cdot b_L}\cdot\frac{C_L}{C_D}\cdot\frac{dm}{m}$

$$s = \frac{V^{n_V+1}}{g\cdot b_L}\cdot\frac{C_L}{C_D}\cdot\ln\left(\frac{m_1}{m_2}\right)$$

Initial mass

Final mass

$$s = \frac{V^{n_V+1}}{g\cdot b_L}\cdot\frac{C_L}{C_{D0}+k\cdot C_L^2}\cdot\ln\left(\frac{m_1}{m_2}\right)$$

Different cases:

1) Piston Engine: $n_V = -1 \rightarrow s \sim \frac{C_L}{C_D} \rightarrow C_L^* = \sqrt{\frac{C_{D0}}{k}}$

max. Distance ε^*

$V^* =$ solving Lift Equation

2) Turbojet Engine: $n_V = 0 \rightarrow s \sim V\cdot\frac{C_L}{C_D}$

V given (easy)

V not given

Excursion:

Turbojet, V not given:

$$s \sim V\frac{C_L}{C_D} = \sqrt{\frac{2mg}{\rho S_{ref}\cdot C_L}}\cdot\frac{C_L}{C_D} = \sqrt{\frac{2mg\,C_L}{\rho S_{ref}}}\cdot\frac{1}{C_D} \sim \underbrace{\boxed{\frac{\sqrt{C_L}}{C_{D0}+k\cdot C_L^2}}}_{f(C_L)}$$

to find optimum C_L: $\frac{\partial f(C_L)}{\partial C_L} = 0$

$$f(C_L) = \frac{\sqrt{C_L}}{C_{D0}+k\cdot C_L^2} = (C_L)^{1/2}\cdot(C_{D0}+k\cdot C_L^2)^{-1}$$

$$\frac{\partial f(C_L)}{\partial C_L} = 0 = \frac{1}{2}\cdot(C_L)^{-1/2}(C_{D0}+k\cdot C_L^2)^{-1} - (C_L^{1/2})(C_{D0}+k\cdot C_L^2)^{-2}\cdot 2kC_L$$

$$= \frac{1/2\cdot(C_L)^{-1/2}\cdot(C_{D0}+k\cdot C_L)}{(C_{D0}+k\cdot C_L^2)^2} - \frac{(C_L)^{1/2}\cdot 2\cdot k\cdot C_L}{(C_{D0}+k\cdot C_L^2)^2}\cdot\frac{C_L}{C_L}$$

$$= \frac{}{(C_{D0} + k\cdot C_L^2)^2} - \frac{}{(C_{D0} + k\cdot C_L^2)^2} \cdot \frac{}{C_L}$$

$$= \frac{1/2\,(C_{D0} + k\cdot C_L)^2}{C_L^{1/2}\cdot (C_{D0}+k\cdot C_L^2)^2} - \frac{2k\cdot C_L^2}{(C_{D0}+k\cdot C_L^2)^2} \cdot \frac{1}{C_L^{1/2}}$$

$$= \frac{1/2\,(C_{D0} + k\cdot C_L^2) - 2kC_L^2}{C_L^{1/2}\,(C_{D0}+k\cdot C_L^2)^2} = \frac{(C_{D0}+k\cdot C_L^2) - 4kC_L^2}{2\,(C_L)^{1/2}\,(C_{D0}+k\cdot C_L^2)^2}$$

$$= \frac{C_{D0} - 3kC_L^2}{2\cdot C_L^{1/2}\,(C_{D0}+kC_L^2)^2} \rightarrow \frac{\partial f(C_L)}{\partial C_L} = 0 \rightarrow 0 = C_{D0} - 3kC_L^2$$

$$\boxed{\Rightarrow C_L = \sqrt{\frac{C_{D0}}{3k}}\ ;}$$

Excursion End and back to Task

Velocity is known: $V = 65\ m/s$

1. Initial Altitude? $\rightarrow$ Solve Lift Equation: $L = m_1\cdot g \rightarrow \frac{1}{2}\rho_1 V S_{ref}\cdot C_L^* = m_1 g$

$$\rightarrow \rho_1 = \frac{2\cdot m_1 g}{V^2\cdot S_{ref}\cdot C_L^*} = \boxed{0.69\ \frac{kg}{m^3};}$$

$$C_L^* = \sqrt{\frac{C_{D0}}{k}}$$

Use ISA Equations:

$$\rho_1 = \rho_{MSL}\left(1 + \frac{\gamma_{TR}}{T}\cdot H_{G1}\right)^{\frac{1}{n-1}}$$

$$H_{G1} = \frac{\left(\left(\frac{\rho_1}{\rho_{MSL}}\right)^{n-1} - 1\right)\cdot T_{MSL}}{\gamma_{TR}} = 5568.7\ m;$$

convert to geometric/geodetic

$$h = \frac{r_E\cdot H_{G1}}{r_E - H_{G1}} = 5573.6\ m;$$

2. Final Altitude:

$$\underset{\text{known}}{s} = \frac{\overset{\text{known}}{V^{n_V+1}}}{\underset{\text{known}}{g\cdot b_c}} \cdot \frac{\overset{\text{known}}{C_L}}{\underset{\text{known by means of sink grad. drag pol.}}{C_D}} \cdot \ln\left(\frac{\overset{\text{given}}{m_1}}{\underset{\text{unknown}}{m_2}}\right)$$

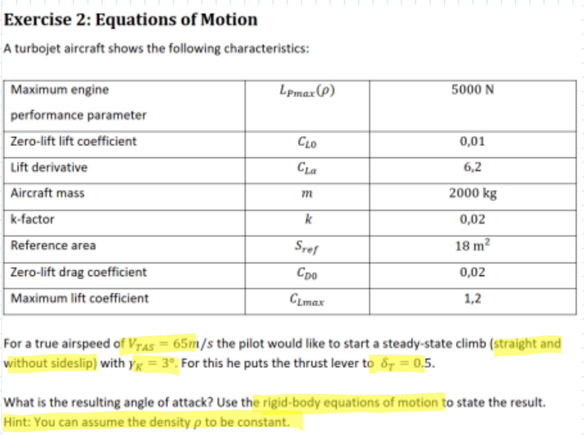
$\rightarrow$ solve for m_2:

$$m_2 = \frac{m_1}{\exp\left(\frac{s\cdot g\cdot b_c}{V^{n_V+1}}\cdot\frac{C_D}{C_L}\right)} = \frac{m_1}{\exp\left(\frac{s\cdot g\cdot b_c}{V^{n_V+1}}\cdot\frac{C_D^*}{C_L^*}\right)} = 1880.6\ kg;$$

solve lift equation with m_2 at point 2

$$L_2 = m_2\cdot g \rightarrow \frac{1}{2}\rho_2 V^2\cdot S_{ref}\cdot C_L^* = m_2\cdot g \rightarrow \rho = 0.65\ kg/m^3$$

$\rightarrow$ ISA $\rightarrow H_G = 6125.4\ m \rightarrow h = 6131.3\ m;$



Exercise 2: Equations of Motion

A turbojet aircraft shows the following characteristics:

Maximum engine performance parameter	$L_{Pmax}(\rho)$	5000 N
Zero-lift lift coefficient	C_{L0}	0,01
Lift derivative	$C_{L\alpha}$	6,2
Aircraft mass	m	2000 kg
k-factor	k	0,02
Reference area	S_{ref}	18 m^2
Zero-lift drag coefficient	C_{D0}	0,02
Maximum lift coefficient	C_{Lmax}	1,2

For a true airspeed of $V_{TAS} = 65m/s$ the pilot would like to start a steady-state climb (straight and without sideslip) with $\gamma_K = 3°$. For this he puts the thrust lever to $\delta_T = 0.5$.

What is the resulting angle of attack? Use the rigid-body equations of motion to state the result.
Hint: You can assume the density ρ to be constant.

Steady state : $\dot V_K^G = \dot\alpha_K = \dot\beta_K \overset{!}{=} 0 = p_K = q_K = r_K$
straight and without sideslip

1) Rigid Body Equations of Motion : No bank and windfree : $K = \bar{K}$

$$\dot V_K^G = \frac{(X_T^G)_K}{m} = \frac{T-D}{m} - g\cdot\sin\gamma = 0$$

$$\dot\alpha_K = \frac{(Z_T^G)_K}{m\cdot V_K^G\cdot\cos\beta_K} + q_K - \tan\beta_K\cdot(p_K\cos\alpha_K + r_K\cdot\sin\alpha_K) = \frac{-L}{m V_K^G \cancelto{=1}{\cos\beta_K}} + \frac{g}{V}\frac{\cancelto{=1}{\cos\mu}\cos\gamma}{\cancelto{=1}{\cos\beta_K}} + \left[\cancelto{=0}{q_K} - \cancelto{=0}{\tan\beta_K}(\cancelto{=0}{p_K}\cdot\cos\alpha_K + \cancelto{=0}{r_K}\cdot\sin\alpha_K)\right] = 0$$

$$\dot\beta_K = \frac{(Y_T^G)_K}{m\cdot V_K^G} + (p_K\cdot\sin\alpha_K - r_K\cdot\cos\alpha_K) = \frac{\cancelto{=0}{Q}}{m\cdot V_K^G} + \frac{g}{V}\cos\gamma\,\cancelto{=0}{\sin\mu} + (\cancelto{=0}{p_K}\cdot\sin\alpha_K - \cancelto{=0}{r_K}\cdot\cos\alpha_K) = 0$$

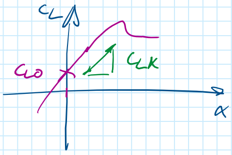
Insert the forces

$$(X_T^G)_K = T - D \boxed{- mg\sin\gamma}$$
$$(Y_T^G)_K = Q \boxed{+ mg\cos\gamma\sin\mu}$$
$$(Z_T^G)_K = -L \boxed{+ mg\cos\mu\cdot\cos\gamma}$$

from $M_{KO}\begin{pmatrix}0\\0\\mg\end{pmatrix}_O$

Resulting Equations:

$$0 = \frac{T-D}{m} - g\sin\gamma_K$$
$$0 = \frac{-L}{mV_K^G} + \frac{g}{V_K^G}\cos\gamma_K$$
$$[0 = \dot\beta_K]$$



Lets find α: $L = \frac{1}{2}\rho V^2 S_{ref}(C_{L0} + C_{L\alpha}\cdot\alpha)$

$$0 = \frac{\rho (V_K^G)^2 S_{ref}(C_{L0} + C_{L\alpha}\cdot\alpha)}{2\,m\,V_K^G} + \frac{g}{V_K^G}\cos\gamma \quad (2)$$

Two Unknowns:

$$0 = \frac{T-D}{m} - g\sin\gamma_K \rightarrow D = T - mg\sin\gamma$$
$$\downarrow$$
$$\frac{T}{T_{ref}} = \left(\frac{V}{V_{ref}}\right)^{n_V}\cdot\left(\frac{\rho}{\rho_{ref}}\right)^{n_\rho} = \boxed{\left(\frac{1}{V_{ref}}\right)^{n_V}\cdot\left(\frac{\rho}{\rho_{ref}}\right)^{n_\rho}}\cdot V^{n_V}$$

$n_V = 0$, $L_P(\rho)$

$$\Rightarrow T = L_P(\rho)\cdot\cancelto{n_V=0}{V^{n_V}} \rightarrow T = L_P(\rho)$$
$$T = L_{P,max}(\rho)\cdot\delta_T$$

given

$$D = 2500N - 2000kg\cdot 9.81\frac{m}{s^2}\cdot\sin\left(3\cdot\frac{\pi}{180}rad\right)$$
$$= 1473.2N = \frac{1}{2}\rho V^2\cdot S_{ref}(C_{D0} + k\cdot C_L^2) \quad (2)$$

Combining Eq. (1) and (2)

$$0 = \frac{-\rho V^2 S_{ref} \cdot C_L}{2\ mV} + \frac{g}{V}\cos\gamma \quad ; D = \frac{1}{2}\rho V^2 S_{ref}\left(C_{D0} + k \cdot C_L^2\right)$$

$$\searrow \quad \swarrow$$

$$\frac{1}{2}\rho V^2 S_{ref} = \frac{D}{C_{D0} + k \cdot C_L^2}$$

$$0 = -D C_L + mg\cos\gamma\left(C_{D0} + k \cdot C_L^2\right)$$

→ Quadratic Equation: $\underbrace{k \cdot mg\cos\gamma}_{a} \cdot C_L^2 \underbrace{- D}_{b} \cdot C_L + \underbrace{mg\cos\gamma\, C_{D0}}_{c} = 0$

$$x_{1,2} = C_{L1,2} = \frac{-b \pm \sqrt{b^2 - 4 \cdot a \cdot c}}{2a}$$

⇒ Results: $C_{L1} = 3.4718$

$$\boxed{C_{L2} = 0.2881} \quad C_{L,max} = 1.2 \ ;$$

$\Rightarrow C_{L2} = 0.2881$;

⇒ Go back to Lift equation: $C_{L2} = C_{L0} + C_{L\alpha} \cdot \alpha$

$$\alpha = \frac{C_{L2} - C_{L0}}{C_{L\alpha}} = 0.0449\,\text{rad} \sim \underline{2.57\,\text{deg}};$$